

COP509 Notebook 2: Extraction, Alignment And Classification

This notebook is the assessed Task 2 notebook. It demonstrates recommendation extraction, response-unit extraction, alignment, classification, confidence metadata, validation evidence and an optional interactive inspection demo.

The notebook reads the final validated outputs from project-relative paths and does not overwrite `final_recommendations_246.json`, `evaluation_predictions.csv`, or any source data.

Google Colab setup

If this notebook is opened through the GitHub Colab link, run the setup cells below first. They clone the full repository so the notebook can access `src/`, `data/`, `outputs/` and the PDF files. Local Jupyter/VS Code users can run the notebook normally.

```
# Colab setup: clone the full repository so src/, data/ and outputs/
are available.
import os
import sys
from pathlib import Path

try:
    import google.colab # type: ignore
    IN_COLAB = True
except ImportError:
    IN_COLAB = False

if IN_COLAB:
    repo_dir = Path("/content/cop509-public-inquiry-nlp")
    if not repo_dir.exists():
        !git clone https://github.com/DanielLawrence04/cop509-public-
inquiry-nlp.git {repo_dir}
    os.chdir(repo_dir)
    print(f"Running from: {Path.cwd()}")
else:
    print("Not running in Colab; using the local repository path.")

# Install required packages when running in Colab.
import sys
from pathlib import Path

try:
    import google.colab # type: ignore
    IN_COLAB = True
```

```

except ImportError:
    IN_COLAB = False

if IN_COLAB:
    !{sys.executable} -m pip install -r requirements.txt
else:
    print("Local environment detected; skipping Colab pip install.")

from __future__ import annotations

import json
import logging
import sys
import warnings
from collections import Counter, defaultdict
from pathlib import Path

import pandas as pd
from IPython.display import Markdown, display

pd.set_option("display.max_colwidth", 140)

# Suppress harmless optional model-library noise while preserving real exceptions.
warnings.filterwarnings("ignore", category=FutureWarning,
    module=r"transformers(\.|$)")
warnings.filterwarnings("ignore", category=FutureWarning,
    module=r"huggingface_hub(\.|$)")
warnings.filterwarnings("ignore", message=r".*resume_download.*",
    category=FutureWarning)
warnings.filterwarnings("ignore", message=r".*Torch was not compiled with flash attention.*")
for logger_name in ("transformers", "sentence_transformers",
    "huggingface_hub"):
    logging.getLogger(logger_name).setLevel(logging.ERROR)
try:
    from transformers.utils import logging as transformers_logging

    transformers_logging.set_verbosity_error()
except Exception:
    pass

def find_project_root(start: Path | None = None) -> Path:
    """Find the project root from the repository layout, not a machine-specific path."""
    start = (start or Path.cwd()).resolve()
    required_dirs = ("src", "data", "outputs")
    coursework_names = ("COP509_Coursework.pdf",
        "COP509_coursework.pdf")

```

```

    for candidate in [start, *start.parents]:
        has_required_dirs = all((candidate / name).is_dir() for name
in required_dirs)
        has_coursework_file = any((candidate / name).exists() for name
in coursework_names)
        if has_required_dirs and (has_coursework_file or (candidate /
"outputs" / "final_recommendations_246.json").exists()):
            return candidate
        raise FileNotFoundError(
            "Could not find the project root. Expected directories: src/,
data/, outputs/ "
            "and either the final export or the COP509 coursework PDF."
        )

PROJECT_ROOT = find_project_root()
DATA_DIR = PROJECT_ROOT / "data"
DATA_RAW_DIR = DATA_DIR / "raw"
OUTPUTS_DIR = PROJECT_ROOT / "outputs"
FINAL_EXPORT_PATH = OUTPUTS_DIR / "final_recommendations_246.json"
EVALUATION_PREDICTIONS_PATH = OUTPUTS_DIR /
"evaluation_predictions.csv"
QA_MATRIX_PATH = DATA_DIR / "ground_truth" / "qa_matrix_queries.json"

# Backward-compatible alias used by older display cells.
RAW_DIR = DATA_RAW_DIR
if str(PROJECT_ROOT) not in sys.path:
    sys.path.insert(0, str(PROJECT_ROOT))

logging.getLogger("src.preprocessing").setLevel(logging.ERROR)
logging.getLogger("src.pdf_loader").setLevel(logging.ERROR)
print(f"Project root detected: {PROJECT_ROOT.name}")
print(f"Raw PDF directory: {DATA_RAW_DIR.relative_to(PROJECT_ROOT)}")
print(f"Final export: {FINAL_EXPORT_PATH.relative_to(PROJECT_ROOT)}")

Project root detected: cop509-coursework
Raw PDF directory: data\raw
Final export: outputs\final_recommendations_246.json

```

1. Coursework Task Overview

Task 2 converts paired report and government-response PDFs into recommendation-level rows. In the current final output, this means **246 recommendation rows across the full 8-pair / 16-PDF corpus**. For each row the project records the extracted recommendation, matched response text, classification label, confidence values and validation metadata.

The implementation is more than a minimal rules demo: it combines preset-aware recommendation extraction, response-unit extraction, structured response matching, classification, confidence scoring, final export validation and automated QA/regression tests used during development.

Full manual labels are not available for all 246 rows. This notebook therefore reports validation checks, row counts, label distributions, confidence summaries, inspected examples and a representative Task 2c expected-vs-actual sample. Accuracy and F1 should only be calculated if a full 246-row manual benchmark is added later.

2. Dataset And Pair Setup

The shared preset map is the source of truth for the full available corpus. It identifies **8 document pairs / 16 PDFs**: coursework-given pairs plus extension pairs used to test the extraction, alignment and classification workflow.

```
from backend.core.presets import PRESETS, validate_preset_files

preset_rows = []
for preset in PRESETS.values():
    try:
        validate_preset_files(preset)
        files_ok, notes = True, ""
    except FileNotFoundError as exc:
        files_ok, notes = False, str(exc)
    special = []
    if preset.allow_response_heading_recommendation_fallback:
        special.append("response-heading fallback")
    if preset.inline_recommendation_chapter_prefix:
        special.append(f"inline prefix {preset.inline_recommendation_chapter_prefix}")
    if preset.select_committee_conclusions_section:
        special.append("select-committee section")
    preset_rows.append({
        "preset_id": preset.id,
        "label": preset.label,
        "dataset_group": preset.dataset_group,
        "policy_pdf": preset.policy_pdf.name,
        "response_pdf": preset.response_pdf.name,
        "files_ok": files_ok,
        "special_extraction": ", ".join(special),
        "notes": notes,
    })

preset_df = pd.DataFrame(preset_rows)
if not preset_df["files_ok"].all():
    display(preset_df)
    raise FileNotFoundError("One or more preset PDFs are missing.")
all_preset_pdf_names = sorted({path.name for preset in
PRESETS.values() for path in (preset.policy_pdf,
preset.response_pdf)})
print(f"Full available preset corpus: {len(PRESETS)} document pairs / {len(all_preset_pdf_names)} PDFs")
print("Task 2 final output is reported across the full 8-pair corpus")
```

```
below.")
display(preset_df)
```

Full available preset corpus: 8 document pairs / 16 PDFs
Task 2 final output is reported across the full 8-pair corpus below.

	preset_id	label
\		
0	behaviour_change	Behaviour Change
1	post_office	Post Office Horizon Inquiry
2	space_economy	The Space Economy
3	covid_inquiry	UK Covid-19 Inquiry Module 1
4	blood_inquiry	Infected Blood Inquiry
5	grenfell_phase2	Grenfell Tower Inquiry – Phase 2
6	covid_inquiry_module2	UK Covid-19 Inquiry Module 2
7	summer_2024_disorder	Police response to the 2024 summer disorder

	dataset_group	policy_pdf	\
0	coursework_given	Behaviour-Change-Report-Recomm.pdf	
1	coursework_given	PostOfficeHorizon-I- Inquiry-Recomm.pdf	
2	coursework_given	TheSpaceEconomyReport.pdf	
3	coursework_given	UK-Covid-19-Inquiry-Module-1-Recomm.pdf	
4	coursework_given	Volume_1-Blood-Inquiry-Recomm.pdf	
5	extra_found	Grenfell-Phase2-Volume7-Recomm.pdf	
6	extra_found	UK-Covid-19-Inquiry-Module-2-Recomm.pdf	
7	extra_found	Summer2024-Disorder-Recomm.pdf	

	response_pdf	files_ok	\
0	Behaviour-Change-Response.pdf	True	
1	PostOfficeHorizon-IT-Inquiry-Response.pdf	True	
2	TheSpaceEconomyResponse.pdf	True	
3	UK-Covid-19_Inquiry_Module_1_Response.pdf	True	
4	Volume_1-Blood-Inquiry-Response.pdf	True	
5	Grenfell-Phase2-Response.pdf	True	
6	UK-Covid-19_Inquiry_Module_2_Response.pdf	True	
7	Summer2024-Disorder-Response.pdf	True	

	special_extraction notes
0	
1	
2	
3	
4	

```
5 response-heading fallback, inline prefix 113
6
7 select-committee section
```

3. Source-Code Mirroring

The notebook imports the same shared project modules used by the final validator. Notebook-local code is limited to display helpers and summary tables.

```
from src.pdf_loader import extract_pages
from src.extraction import extract_recommendations
from src.response_units import extract_response_units
from src.alignment import match_recommendations_to_response_units
from src.classification import classify_response,
classify_with_confidence, normalize_label
from scripts.validate_recommendation_export import (
    classification_failures,
    metadata_failures,
    recommendation_cleanup_failures,
    response_leakage_failures,
)

TASK2_LABELS = ["accepted", "partial", "rejected", "not_addressed"]
LABEL_DISPLAY = {
    "accepted": "accepted",
    "partial": "partially accepted",
    "partially_accepted": "partially accepted",
    "rejected": "rejected",
    "not_addressed": "not addressed",
}
print("Loaded source modules for loading, extraction, response units,
alignment and classification.")
```

Loaded source modules for loading, extraction, response units,
alignment and classification.

4. Final Validated Export Baseline

The selected evidence baseline is the canonical project-relative file `outputs/final_recommendations_246.json`. The supporting classified export is `outputs/evaluation_predictions.csv`. The cell below reads the JSON baseline, checks the expected row counts and leaves the source outputs unchanged.

```
EXPECTED_PAIR_COUNTS = {
    "behaviour_change": 33,
    "post_office": 19,
    "space_economy": 40,
    "covid_inquiry": 10,
```

```

        "blood_inquiry": 58,
        "grenfell_phase2": 58,
        "covid_inquiry_module2": 19,
        "summer_2024_disorder": 9,
    }
    EXPECTED_CLASS_COUNTS = {"accepted": 104, "partial": 139, "rejected":
1, "not_addressed": 2}
    FINAL_EXPORT_CANDIDATES = [FINAL_EXPORT_PATH]

    def load_export(path: Path) -> list[dict]:
        payload = json.loads(path.read_text(encoding="utf-8"))
        rows = payload.get("recommendations", payload) if
isinstance(payload, dict) else payload
        if not isinstance(rows, list):
            raise ValueError(f"{path} does not contain a recommendation
list")
        return rows

    selected_export_path, final_rows = None, None
    for candidate in FINAL_EXPORT_CANDIDATES:
        if not candidate.exists():
            continue
        rows = load_export(candidate)
        if (
            len(rows) == 246
            and dict(Counter(str(r.get("document_pair")) for r in rows))
== EXPECTED_PAIR_COUNTS
            and dict(Counter(str(r.get("classification")) for r in rows))
== EXPECTED_CLASS_COUNTS
        ):
            selected_export_path, final_rows = candidate, rows
            break

    if final_rows is None:
        found = [str(path) for path in FINAL_EXPORT_CANDIDATES if
path.exists()]
        raise FileNotFoundError("No validated 246-row final export found.
Expected: " + str(FINAL_EXPORT_PATH.relative_to(PROJECT_ROOT)))

    final_df = pd.json_normalize(final_rows, sep=".")
    print("Selected final export evidence baseline:")
    print(selected_export_path.relative_to(PROJECT_ROOT))
    print("Canonical evidence name used in this notebook:
final_recommendations_246.json")
    print(f"Rows loaded: {len(final_df)}")

    Selected final export evidence baseline:
    outputs\final_recommendations_246.json
    Canonical evidence name used in this notebook:

```

```
final_recommendations_246.json
Rows loaded: 246
```

5. Final Full Export Summary

The compact export label `partial` means partially accepted.

```
pair_summary = pd.DataFrame({
    "document_pair": list(EXPECTED_PAIR_COUNTS),
    "expected_rows": list(EXPECTED_PAIR_COUNTS.values()),
    "actual_rows": [int((final_df["document_pair"] == pair).sum()) for
pair in EXPECTED_PAIR_COUNTS],
})
pair_summary["matches_expected"] = pair_summary["expected_rows"] ==
pair_summary["actual_rows"]

class_counts = final_df["classification"].value_counts().to_dict()
class_summary = pd.DataFrame({
    "export_label": TASK2_LABELS,
    "display_label": [LABEL_DISPLAY[label] for label in TASK2_LABELS],
    "expected_rows": [EXPECTED_CLASS_COUNTS[label] for label in
TASK2_LABELS],
    "actual_rows": [int(class_counts.get(label, 0)) for label in
TASK2_LABELS],
})
class_summary["percentage"] = (class_summary["actual_rows"] /
len(final_df) * 100).round(1)
class_summary["matches_expected"] = class_summary["expected_rows"] ==
class_summary["actual_rows"]

print(f"Final row count: {len(final_df)}")
display(pair_summary)
display(class_summary)
```

Final row count: 246

	document_pair	expected_rows	actual_rows	matches_expected
0	behaviour_change	33	33	True
1	post_office	19	19	True
2	space_economy	40	40	True
3	covid_inquiry	10	10	True
4	blood_inquiry	58	58	True
5	grenfell_phase2	58	58	True
6	covid_inquiry_module2	19	19	True
7	summer_2024_disorder	9	9	True

	export_label	display_label	expected_rows	actual_rows
percentage \				
0	accepted	accepted	104	104

42.3					
1	partial	partially accepted		139	139
56.5					
2	rejected	rejected		1	1
0.4					
3	not_addressed	not addressed		2	2
0.8					

	matches_expected
0	True
1	True
2	True
3	True

6. Confidence, Metadata And Validation Checks

The final export includes overall confidence plus debug fields for alignment confidence, lexical similarity, classification confidence, classifier method, alignment method and confidence factors.

```
confidence_summary = pd.DataFrame([
    {"metric": "overall confidence", "mean":
final_df["confidence"].mean(), "min": final_df["confidence"].min(),
"max": final_df["confidence"].max()},
    {"metric": "alignment confidence", "mean":
final_df["debug.alignment_confidence"].mean(), "min":
final_df["debug.alignment_confidence"].min(), "max":
final_df["debug.alignment_confidence"].max()},
    {"metric": "classification confidence", "mean":
final_df["debug.classification_confidence"].mean(), "min":
final_df["debug.classification_confidence"].min(), "max":
final_df["debug.classification_confidence"].max()},
]).round(3)
method_summary =
final_df["debug.alignment_method"].value_counts(dropna=False).rename_
axis("alignment_method").reset_index(name="rows")
classifier_summary =
final_df["debug.classifier_method"].value_counts(dropna=False).rename_
axis("classifier_method").reset_index(name="rows")
display(confidence_summary)
display(method_summary)
display(classifier_summary)

def duplicate_id_failures(rows: list[dict]) -> dict[str, list[str]]:
    by_pair = defaultdict(list)
    for row in rows:
        by_pair[str(row.get("document_pair"))].append(str(row.get("id")))
```

```

    return {pair: sorted([item for item, count in Counter(ids).items()
if count > 1]) for pair, ids in by_pair.items() if any(count > 1 for
count in Counter(ids).values())}

def blank_id_failures(rows: list[dict]) -> list[tuple[str, str]]:
    return [(str(r.get("document_pair")), str(r.get("id"))) for r in
rows if r.get("id") is None or str(r.get("id")).strip() == ""]

validation_df = pd.DataFrame([
    {"check": "row count", "status": "ok" if len(final_rows) == 246
else "failed", "details": len(final_rows)},
    {"check": "document-pair counts", "status": "ok" if
pair_summary["matches_expected"].all() else "failed", "details": "see
pair summary"},
    {"check": "classification distribution", "status": "ok" if
class_summary["matches_expected"].all() else "failed", "details": "see
class summary"},
    {"check": "duplicate IDs", "status": "ok" if not
duplicate_id_failures(final_rows) else "failed", "details":
duplicate_id_failures(final_rows) or "none"},
    {"check": "blank IDs", "status": "ok" if not
blank_id_failures(final_rows) else "failed", "details":
blank_id_failures(final_rows) or "none"},
    {"check": "response leakage", "status": "ok" if not
response_leakage_failures(final_rows) else "failed", "details":
response_leakage_failures(final_rows) or "none"},
    {"check": "recommendation cleanup", "status": "ok" if not
recommendation_cleanup_failures(final_rows) else "failed", "details":
recommendation_cleanup_failures(final_rows) or "none"},
    {"check": "metadata completeness", "status": "ok" if not
metadata_failures(final_rows) else "failed", "details":
metadata_failures(final_rows) or "none"},
    {"check": "classification corrections", "status": "ok" if not
classification_failures(final_rows) else "failed", "details":
classification_failures(final_rows) or "none"},
])
display(validation_df)
if not (validation_df["status"] == "ok").all():
    raise AssertionError("One or more final export validation checks
failed.")

```

	metric	mean	min	max
0	overall confidence	0.935	0.8	0.95
1	alignment confidence	0.938	0.9	0.95
2	classification confidence	0.787	0.6	0.92

	alignment_method	rows
0	exact_label	148
1	structured_grouped	52

2	structure	30	
3	structured_ordinal	10	
4	structured_section	6	
	classifier_method	rows	
0	rule_based	246	
		check status	details
0	row count	ok	246
1	document-pair counts	ok	see pair summary
2	classification distribution	ok	see class summary
3	duplicate IDs	ok	none
4	blank IDs	ok	none
5	response leakage	ok	none
6	recommendation cleanup	ok	none
7	metadata completeness	ok	none
8	classification corrections	ok	none

7. Live Pipeline Demonstration On One Pair

The final export above is loaded read-only. This section runs the shared project modules on one representative pair so the extraction, response-unit and alignment stages are visible without changing final outputs.

```

DEMO_PRESET_ID = "behaviour_change"
demo_preset = PRESETS[DEMO_PRESET_ID]
policy_pages = extract_pages(demo_preset.policy_pdf)
response_pages = extract_pages(demo_preset.response_pdf)
demo_recommendations = extract_recommendations(
    policy_pages,
    response_pages_fallback=(response_pages if
demo_preset.allow_response_heading_recommendation_fallback else None),

inline_chapter_prefix=demo_preset.inline_recommendation_chapter_prefix
,

select_committee_section=demo_preset.select_committee_conclusions_section,
)
demo_response_units = extract_response_units(response_pages)
demo_alignments =
match_recommendations_to_response_units(demo_recommendations,
demo_response_units, top_k=3, similarity_threshold=0.05)
print(f"Demo preset: {DEMO_PRESET_ID} - {demo_preset.label}")
print(f"Policy pages: {len(policy_pages)}")
print(f"Response pages: {len(response_pages)}")
print(f"Recommendations extracted: {len(demo_recommendations)}")
print(f"Response units extracted: {len(demo_response_units)}")
print(f"Alignment candidates returned: {len(demo_alignments)}")

```

Demo preset: behaviour_change - Behaviour Change
Policy pages: 109
Response pages: 18
Recommendations extracted: 33
Response units extracted: 22
Alignment candidates returned: 99

8. Recommendation Extraction And Response Matching

Recommendation extraction records method and confidence. Response-unit matching returns alignment method, similarity and alignment confidence.

```
demo_rec_df = pd.DataFrame(demo_recommendations)
display(demo_rec_df[["extraction_method"].value_counts().rename_axis("e
xtraction_method").reset_index(name="rows"))
display(demo_rec_df[["rec_id", "item_label", "page_number",
"extraction_method", "confidence", "text"]].assign(text=lambda df:
df["text"].str.slice(0, 180) + "...").head(8))

unit_df = pd.DataFrame(demo_response_units)
if not unit_df.empty:
    unit_df =
unit_df.assign(response_preview=unit_df["response_text"].fillna("").st
r.slice(0, 160) + "...")
    display(unit_df[["recommendation_labels", "page_start",
"page_end", "extraction_confidence", "boundary_reason",
"response_preview"]].head(8))

demo_align_df = pd.DataFrame(demo_alignments)
if not demo_align_df.empty:
    display(demo_align_df.groupby("match_method",
as_index=False).agg(rows=("rec_id", "count"),
mean_similarity=("similarity", "mean"),
mean_alignment_confidence=("alignment_confidence", "mean")).round(3))
    display(demo_align_df[["rec_id", "match_method", "similarity",
"alignment_confidence", "page_number",
"matched_text"]].assign(matched_text=lambda df:
df["matched_text"].str.slice(0, 180) + "...").head(8))
```

	extraction_method	rows
0	structured section	33

	rec_id	item_label	page_number	extraction_method	confidence	\
0	0	8.1	67	structured section	0.86	
1	1	8.2	67	structured section	0.86	
2	2	8.3	67	structured section	0.86	
3	3	8.4	67	structured section	0.92	
4	4	8.5	67	structured section	0.92	
5	5	8.6	68	structured section	0.92	

6	6	8.7	68	structured section	0.92
7	7	8.8	68	structured section	0.92

text

0 The idea of the Government intervening to change people's behaviour will often be controversial, and so it is important that ministers a...

1 There is a lack of applied research at a population level to support specific interventions to change the behaviour of large groups of p...

2 We acknowledge that there will be occasions when it is legitimate for a government not to implement behaviour change interventions for w...

3 We agree with the principle, stated in the Government's Principles of Scientific Advice , that ministers should explain publicly their r...

4 We urge ministers to consult their departmental Chief Scientific Advisers about whether the amount of money spent on applied behaviour c...

5 We recommend that, at the earliest opportunity, the Government appoint a Chief Social Scientist who reports to the Government Chief Scie...

6 We further recommend that the Government consider whether existing mechanisms for the provision of social scientific advice, in particul...

7 Departmental Chief Scientific Advisers, whether or not they have experience of the sciences of human behaviour, should be responsible fo...

	recommendation_labels	page_start	page_end	
	extraction_confidence \			
0	[8.1]	3	3	0.95
1	[8.2]	3	3	0.95
2	[8.3, 8.4]	4	4	0.75
3	[8.5]	4	4	0.95
4	[8.6, 8.7]	4	5	0.75
5	[8.8, 8.9]	5	6	0.75
6	[8.10, 8.11]	6	6	0.75
7	[8.12, 8.13]	6	7	0.75

boundary_reason \

```

0      next_heading
1      next_heading
2  multi_label_block
3      next_heading
4  multi_label_block
5  multi_label_block
6  multi_label_block
7  multi_label_block

```

response_preview

0 The Government agrees with the Committee's conclusion. It is important that behavioural interventions draw from the available evidence b...

1 The Government agrees with the Committee's conclusion. As with all policy decisions, we believe that it is important that the Government...

2 The Government agrees with the Committee's conclusion. As with all policy decisions and in line with the GCSA Guidelines on the use of s...

3 The Government agrees with the need to ensure that Ministers are able to consult departmental experts when considering the amount of spe...

4 It is essential that policy making is informed by high quality evidence drawing upon the full range of analytical disciplines. The Gover...

5 The Government agrees that the best expertise and advice from across all the relevant disciplines are brought to bear on decisions in th...

6 The Cabinet Office has already published a paper, jointly with the Institute for Government, which sets out an evidence-based framework ...

7 We agree that it is important to ensure that effective mechanisms exist for sharing and disseminating knowledge about behavioural interv...

	match_method	rows	mean_similarity	mean_alignment_confidence
0	exact_label	33	0.950	0.950
1	semantic	66	0.127	0.127

	rec_id	match_method	similarity	alignment_confidence	page_number
0	0	exact_label	0.950000	0.950000	3
1	0	semantic	0.128381	0.128381	11
2	0	semantic	0.126074	0.126074	10
3	1	exact_label	0.950000	0.950000	3

4	1	semantic	0.130277	0.130277	8
5	1	semantic	0.123090	0.123090	18
6	2	exact_label	0.950000	0.950000	4
7	2	semantic	0.131071	0.131071	11

matched_text

0 The Government agrees with the Committee's conclusion. It is important that behavioural interventions draw from the available evidence b...

1 The Government agrees with the Committee's recommendation. However, we would emphasise that Change4Life presents a number of challenges ...

2 The Government agrees that it is important for local authorities to have the ability to tailor their interventions to local populations....

3 The Government agrees with the Committee's conclusion. As with all policy decisions, we believe that it is important that the Government...

4 As the Committee has acknowledged, collaboration between Government, public\health, commercial and voluntary organisations presents a u...

5 The Department for Transport drew local authorities' attention to a wide range of research and evaluation evidence when the Local Sustai...

6 The Government agrees with the Committee's conclusion. As with all policy decisions and in line with the GCSA Guidelines on the use of s...

7 The Government agrees with the Committee's recommendation. However, we would emphasise that Change4Life presents a number of challenges ...

9. Classification Labels

The classifier returns `accepted`, `partially_accepted`, `rejected` or `not_addressed`. The final export normalises `partially_accepted` to the compact `partial` value.

```
alignments_by_rec = defaultdict(list)
for alignment in demo_alignments:

    alignments_by_rec[int(alignment["rec_id"])].append(dict(alignment))

classified_demo_rows = []
for rec in demo_recommendations:
    rec_matches = sorted(alignments_by_rec.get(int(rec["rec_id"]),
```

```

[[]], key=lambda item: float(item.get("similarity", 0.0)),
reverse=True)
    best = rec_matches[0] if rec_matches else None
    if best:
        raw_label, class_conf =
classify_with_confidence(str(best.get("matched_text") or ""))
        label = normalize_label(raw_label)
    else:
        label, class_conf = "not_addressed", 0.0
    export_label = "partial" if label == "partially_accepted" else
label
    classified_demo_rows.append({
        "item_label": rec.get("item_label"),
        "classification": LABEL_DISPLAY.get(export_label,
export_label),
        "classification_confidence": round(float(class_conf), 3),
        "alignment_confidence": round(float((best or
{}).get("alignment_confidence", 0.0) or 0.0), 3),
        "match_method": (best or {}).get("match_method"),
        "recommendation_preview": str(rec.get("text", ""))[:120] +
"...",
        "response_preview": str((best or {}).get("matched_text", ""))
[:120] + "...",
    })
classified_demo_df = pd.DataFrame(classified_demo_rows)
display(classified_demo_df.head(10))
display(classified_demo_df["classification"].value_counts().rename_axi
s("classification").reset_index(name="rows"))

```

	item_label	classification	classification_confidence \
0	8.1	accepted	0.78
1	8.2	accepted	0.85
2	8.3	accepted	0.75
3	8.4	accepted	0.75
4	8.5	accepted	0.85
5	8.6	partially accepted	0.70
6	8.7	partially accepted	0.70
7	8.8	accepted	0.92
8	8.9	accepted	0.92
9	8.10	accepted	0.85

	alignment_confidence	match_method \
0	0.95	exact_label
1	0.95	exact_label
2	0.95	exact_label
3	0.95	exact_label
4	0.95	exact_label
5	0.95	exact_label
6	0.95	exact_label
7	0.95	exact_label

8	0.95	exact_label
9	0.95	exact_label

recommendation_preview \

0 The idea of the Government intervening to change people's behaviour will often be controversial, and so it is important ...

1 There is a lack of applied research at a population level to support specific interventions to change the behaviour of l...

2 We acknowledge that there will be occasions when it is legitimate for a government not to implement behaviour change int...

3 We agree with the principle, stated in the Government's Principles of Scientific Advice , that ministers should explain ...

4 We urge ministers to consult their departmental Chief Scientific Advisers about whether the amount of money spent on app...

5 We recommend that, at the earliest opportunity, the Government appoint a Chief Social Scientist who reports to the Gover...

6 We further recommend that the Government consider whether existing mechanisms for the provision of social scientific adv...

7 Departmental Chief Scientific Advisers, whether or not they have experience of the sciences of human behaviour, should b...

8 We recommend that the Cabinet Secretary, in consultation with the Government Chief Scientific Adviser and Chief Social S...

9 We recommend that the Cabinet Office, in consultation with the Chief Social Scientist, once appointed, consider how to c...

response_preview

0 The Government agrees with the Committee's conclusion. It is important that behavioural interventions draw from the avai...

1 The Government agrees with the Committee's conclusion. As with all policy decisions, we believe that it is important tha...

2 The Government agrees with the Committee's conclusion. As with all policy decisions and in line with the GCSA Guidelines...

3 The Government agrees with the Committee's conclusion. As with all policy decisions and in line with the GCSA Guidelines...

4 The Government agrees with the need to ensure that Ministers are able to consult departmental experts when considering t...

5 It is essential that policy making is informed by high quality evidence drawing upon the full range of analytical discip...

6 It is essential that policy making is informed by high quality evidence drawing upon the full range of analytical discip...

7 The Government agrees that the best expertise and advice from across all the relevant disciplines are brought to bear on...

8 The Government agrees that the best expertise and advice from across all the relevant disciplines are brought to bear on...

9 The Cabinet Office has already published a paper, jointly with the Institute for Government, which sets out an evidence-...

	classification	rows
0	partially accepted	19
1	accepted	14

10. Task 2c Evaluation Note

A full manual ground-truth label set is not available for all 246 final rows. For that reason, this notebook does not report overall accuracy, precision, recall or F1 for Task 2. Instead, it reports final row counts, label distributions, validation checks, confidence summaries, inspected examples and the representative expected-vs-actual Task 2c sample below.

If a complete 246-row manual benchmark is added later, accuracy and F1 can be calculated against that benchmark. The existing `labels.json` file is not a full-row benchmark and is not used for that purpose.

Task 2c Evaluation: Expected vs Actual Sample

A full 246-row manual benchmark is not available, so this section uses a representative manually reviewed sample across document pairs, alignment patterns and classification labels. It is intended to address the Task 2c expected-vs-actual requirement without claiming full-corpus accuracy, precision, recall or F1.

The sample covers accepted, partial, rejected and not addressed outcomes, including easy exact matches, grouped responses, numbering/hard cases and known limitations that should remain in future regression checks.

```
task2c_sample_spec = [
    {
        "document_pair": "behaviour_change",
        "recommendation_id": "8.1",
        "expected_response_summary": "Government agrees that
behaviour-change interventions need an explainable evidence base.",
        "system_response_summary": "Matched response says the
Government agrees and explains use of evidence for behaviour-change
policy.",
        "expected_classification": "accepted",
        "alignment_match": "yes",
        "comment": "Easy exact-numbered match with clear agreement
language.",
        "suggested_fix_if_error": "None - correct in sample.",
    },
    {
        "document_pair": "behaviour_change",
        "recommendation_id": "8.25",
        "expected_response_summary": "Government disagrees with the
committee's view but gives an alternative position on responsibility
pledges.",
        "system_response_summary": "Matched response says the
Government does not agree with the committee's view and explains its
```

```

alternative position.",
    "expected_classification": "partial",
    "alignment_match": "yes",
    "comment": "Mixed disagreement plus explanation is treated as
partial rather than simple rejection.",
    "suggested_fix_if_error": "Keep as a regression case for
disagreement-with-alternative wording.",
},
{
    "document_pair": "behaviour_change",
    "recommendation_id": "8.33",
    "expected_response_summary": "Government does not set car-use
carbon targets but describes wider carbon-reduction policy.",
    "system_response_summary": "Matched response rejects specific
targets while referring to broader economy-wide carbon-reduction
targets.",
    "expected_classification": "partial",
    "alignment_match": "partial",
    "comment": "Known limitation: nearby repeated response
headings make exact numbering alone fragile, although the final row is
usable.",
    "suggested_fix_if_error": "Add stronger heading disambiguation
and keep this as a hard-case regression test.",
},
{
    "document_pair": "post_office",
    "recommendation_id": "4",
    "expected_response_summary": "DBT broadly accepts legal-
support funding but limits scope and capacity.",
    "system_response_summary": "Matched response says DBT broadly
accepts and explains limits on funded legal advice.",
    "expected_classification": "partial",
    "alignment_match": "yes",
    "comment": "Clear partial acceptance language with
qualifications.",
    "suggested_fix_if_error": "None - correct in sample.",
},
{
    "document_pair": "post_office",
    "recommendation_id": "13",
    "expected_response_summary": "DBT rejects closing the Dispute
Resolution Procedure for current claimants.",
    "system_response_summary": "Matched response explicitly says
DBT rejects the recommendation.",
    "expected_classification": "rejected",
    "alignment_match": "yes",
    "comment": "Explicit rejection example.",
    "suggested_fix_if_error": "None - correct in sample.",
},

```

```

{
  "document_pair": "space_economy",
  "recommendation_id": "23",
  "expected_response_summary": "Government explores support for
Earth observation collaboration rather than committing to the full
recommendation.",
  "system_response_summary": "Matched response discusses
exploring opportunities to support the UK EO sector and
collaboration.",
  "expected_classification": "partial",
  "alignment_match": "yes",
  "comment": "Structured response match with exploratory
commitment language.",
  "suggested_fix_if_error": "None - correct in sample.",
},
{
  "document_pair": "space_economy",
  "recommendation_id": "30",
  "expected_response_summary": "Government notes space is
already critical infrastructure and describes access-to-space
policy.",
  "system_response_summary": "Matched response states space is
already designated as Critical National Infrastructure and explains
the policy context.",
  "expected_classification": "partial",
  "alignment_match": "yes",
  "comment": "Hard case where section-heading cleanup avoids
recommendation-text leakage.",
  "suggested_fix_if_error": "Keep cleanup checks for trailing
headings in future exports.",
},
{
  "document_pair": "space_economy",
  "recommendation_id": "69",
  "expected_response_summary": "Government describes
international space-agency agreements but does not provide the
requested complete annual publication.",
  "system_response_summary": "Matched response lists
international partnership agreements and explains current practice.",
  "expected_classification": "partial",
  "alignment_match": "yes",
  "comment": "Structured ordinal response match.",
  "suggested_fix_if_error": "None - correct in sample.",
},
{
  "document_pair": "covid_inquiry",
  "recommendation_id": "2",
  "expected_response_summary": "Government agrees to a greater
Cabinet Office role but retains the lead department model.",

```

```

        "system_response_summary": "Matched response agrees with more
Cabinet Office leadership while keeping the Lead Government Department
model.",
        "expected_classification": "partial",
        "alignment_match": "yes",
        "comment": "Good example of partial implementation against a
multi-part recommendation.",
        "suggested_fix_if_error": "None - correct in sample.",
    },
    {
        "document_pair": "covid_inquiry",
        "recommendation_id": "8",
        "expected_response_summary": "Government agrees with
publishing reports on preparedness and resilience.",
        "system_response_summary": "Matched response agrees with
transparency and public understanding of risks and mitigation.",
        "expected_classification": "accepted",
        "alignment_match": "yes",
        "comment": "Exact label match with clear agreement language.",
        "suggested_fix_if_error": "None - correct in sample.",
    },
    {
        "document_pair": "blood_inquiry",
        "recommendation_id": "4a i",
        "expected_response_summary": "UK nations accept introducing or
supporting statutory candour duties.",
        "system_response_summary": "Matched grouped response says the
recommendation is accepted in full by relevant administrations.",
        "expected_classification": "accepted",
        "alignment_match": "yes",
        "comment": "Grouped/nested response case; label preservation
is important.",
        "suggested_fix_if_error": "Keep grouped-response unit tests
for nested labels.",
    },
    {
        "document_pair": "blood_inquiry",
        "recommendation_id": "4c ii",
        "expected_response_summary": "Governments accept aspects in
principle/full but leave implementation detail conditional.",
        "system_response_summary": "Matched grouped response combines
accepted-in-principle and accepted-in-full wording across
administrations.",
        "expected_classification": "partial",
        "alignment_match": "yes",
        "comment": "Grouped response with mixed commitments across
jurisdictions.",
        "suggested_fix_if_error": "Review jurisdiction-specific
sublabels if a finer benchmark is added.",
    },

```

```

    },
    {
      "document_pair": "grenfell_phase2",
      "recommendation_id": "31",
      "expected_response_summary": "Inspectorate accepts
inspection/reporting recommendation for London Fire Brigade.",
      "system_response_summary": "Matched response says HMICFRS
accepts the recommendation and describes inspection work.",
      "expected_classification": "accepted",
      "alignment_match": "yes",
      "comment": "Clear accepted response in a long inquiry
document.",
      "suggested_fix_if_error": "None - correct in sample.",
    },
    {
      "document_pair": "grenfell_phase2",
      "recommendation_id": "50",
      "expected_response_summary": "Government supports the local-
authority recommendation but frames delivery through guidance and
local constraints.",
      "system_response_summary": "Matched response supports the
recommendation and notes local-authority implementation constraints.",
      "expected_classification": "partial",
      "alignment_match": "yes",
      "comment": "Qualified support is correctly not treated as full
acceptance.",
      "suggested_fix_if_error": "None - correct in sample.",
    },
    {
      "document_pair": "grenfell_phase2",
      "recommendation_id": "58",
      "expected_response_summary": "Government accepts reconsidering
the LGA Guide advice and says new guidance will be published.",
      "system_response_summary": "Matched response accepts the
recommendation and notes the old advice was redacted and new guidance
is planned.",
      "expected_classification": "accepted",
      "alignment_match": "yes",
      "comment": "Hard case from embedded inquiry recommendations;
final row is correctly aligned.",
      "suggested_fix_if_error": "Keep inline-recommendation
extraction checks for Grenfell-style prose.",
    },
    {
      "document_pair": "covid_inquiry_module2",
      "recommendation_id": "1",
      "expected_response_summary": "UK government states the
recommendation is for Northern Ireland rather than for UK government
response.",
    }
  ]

```

```

        "system_response_summary": "Matched response says this
recommendation is not for the UK government to respond to.",
        "expected_classification": "not_addressed",
        "alignment_match": "yes",
        "comment": "Not-addressed example caused by responsibility
outside UK government response scope.",
        "suggested_fix_if_error": "None - correct in sample.",
    },
    {
        "document_pair": "covid_inquiry_module2",
        "recommendation_id": "13",
        "expected_response_summary": "UK government again states the
Northern Ireland recommendation is not for it to answer.",
        "system_response_summary": "Matched response says this
recommendation is not for the UK government to respond to.",
        "expected_classification": "not_addressed",
        "alignment_match": "yes",
        "comment": "Second not-addressed example with exact
numbering.",
        "suggested_fix_if_error": "None - correct in sample.",
    },
    {
        "document_pair": "summer_2024_disorder",
        "recommendation_id": "1",
        "expected_response_summary": "CPS/Government response
describes media protocol work but does not fully satisfy the update
cadence requested.",
        "system_response_summary": "Matched response discusses the
media protocol and information-sharing arrangements after Southport.",
        "expected_classification": "partial",
        "alignment_match": "yes",
        "comment": "Select-committee extraction case; conclusion-only
items are excluded upstream.",
        "suggested_fix_if_error": "Keep select-committee section-
boundary checks for future documents.",
    },
]

```

```

def _safe_text_column(df: pd.DataFrame, column: str) -> pd.Series:
    if column in df.columns:
        return df[column].astype(str)
    return pd.Series([""] * len(df), index=df.index, dtype="string")

def _lookup_final_row(document_pair: str, recommendation_id: str) -> dict:
    pair_col = _safe_text_column(final_df, "document_pair")
    id_col = _safe_text_column(final_df, "id")
    matches = final_df[(pair_col == str(document_pair)) & (id_col ==
str(recommendation_id))]

```

```

    if matches.empty:
        return {}
    return matches.iloc[0].to_dict()

task2c_rows = []
for item in task2c_sample_spec:
    actual = _lookup_final_row(item["document_pair"],
item["recommendation_id"])
    actual_classification = str(actual.get("classification",
"missing"))
    expected_classification = item["expected_classification"]
    task2c_rows.append({
        "document_pair": item["document_pair"],
        "recommendation_id": item["recommendation_id"],
        "expected_response_summary":
item["expected_response_summary"],
        "system_response_summary": item["system_response_summary"],
        "expected_classification": expected_classification,
        "actual_classification": actual_classification,
        "alignment_match": item["alignment_match"],
        "classification_match": actual_classification ==
expected_classification,
        "comment": item["comment"],
        "suggested_fix_if_error": item["suggested_fix_if_error"],
    })

task2c_eval_df = pd.DataFrame(task2c_rows)
task2c_columns = [
    "document_pair",
    "recommendation_id",
    "expected_response_summary",
    "system_response_summary",
    "expected_classification",
    "actual_classification",
    "alignment_match",
    "classification_match",
    "comment",
    "suggested_fix_if_error",
]
display(task2c_eval_df[task2c_columns])

label_coverage =
task2c_eval_df["expected_classification"].value_counts().rename_axis("
label").reset_index(name="sample_rows")
display(label_coverage)
print(f"Task 2c representative sample rows: {len(task2c_eval_df)}")
print("Full-corpus accuracy/F1 is not reported because a complete
manual benchmark is unavailable.")

```

	document_pair	recommendation_id	\
0	behaviour_change	8.1	
1	behaviour_change	8.25	
2	behaviour_change	8.33	
3	post_office	4	
4	post_office	13	
5	space_economy	23	
6	space_economy	30	
7	space_economy	69	
8	covid_inquiry	2	
9	covid_inquiry	8	
10	blood_inquiry	4a	i
11	blood_inquiry	4c	ii
12	grenfell_phase2	31	
13	grenfell_phase2	50	
14	grenfell_phase2	58	
15	covid_inquiry_module2	1	
16	covid_inquiry_module2	13	
17	summer_2024_disorder	1	

expected_response_summary \

0 Government agrees that behaviour-change interventions need an explainable evidence base.

1 Government disagrees with the committee's view but gives an alternative position on responsibility pledges.

2 Government does not set car-use carbon targets but describes wider carbon-reduction policy.

3 DBT broadly accepts legal-support funding but limits scope and capacity.

4 DBT rejects closing the Dispute Resolution Procedure for current claimants.

5 Government explores support for Earth observation collaboration rather than committing to the full recommendation.

6 Government notes space is already critical infrastructure and describes access-to-space policy.

7 Government describes international space-agency agreements but does not provide the requested complete annual publication.

8 Government agrees to a greater Cabinet Office role but retains the lead department model.

9 Government agrees with publishing reports on preparedness and resilience.

10 UK nations accept introducing or supporting statutory candour duties.

11 Governments accept aspects in principle/full but leave implementation detail conditional.

12 Inspectorate accepts inspection/reporting recommendation for London Fire Brigade.

13 Government supports the local-authority recommendation but frames delivery through guidance and local constraints.

14 Government accepts reconsidering the LGA Guide advice and says new guidance will be published.
 15 UK government states the recommendation is for Northern Ireland rather than for UK government response.
 16 UK government again states the Northern Ireland recommendation is not for it to answer.
 17 CPS/Government response describes media protocol work but does not fully satisfy the update cadence requested.

system_response_summary \

0 Matched response says the Government agrees and explains use of evidence for behaviour-change policy.
 1 Matched response says the Government does not agree with the committee's view and explains its alternative position.
 2 Matched response rejects specific targets while referring to broader economy-wide carbon-reduction targets.
 3 Matched response says DBT broadly accepts and explains limits on funded legal advice.
 4 Matched response explicitly says DBT rejects the recommendation.
 5 Matched response discusses exploring opportunities to support the UK EO sector and collaboration.
 6 Matched response states space is already designated as Critical National Infrastructure and explains the policy context.
 7 Matched response lists international partnership agreements and explains current practice.
 8 Matched response agrees with more Cabinet Office leadership while keeping the Lead Government Department model.
 9 Matched response agrees with transparency and public understanding of risks and mitigation.
 10 Matched grouped response says the recommendation is accepted in full by relevant administrations.
 11 Matched grouped response combines accepted-in-principle and accepted-in-full wording across administrations.
 12 Matched response says HMICFRS accepts the recommendation and describes inspection work.
 13 Matched response supports the recommendation and notes local-authority implementation constraints.
 14 Matched response accepts the recommendation and notes the old advice was redacted and new guidance is planned.
 15 Matched response says this recommendation is not for the UK government to respond to.
 16 Matched response says this recommendation is not for the UK government to respond to.
 17 Matched response discusses the media protocol and information-sharing arrangements after Southport.

	expected_classification	actual_classification	alignment_match	\
0	accepted	accepted	yes	

1	partial	partial	yes
2	partial	partial	partial
3	partial	partial	yes
4	rejected	rejected	yes
5	partial	partial	yes
6	partial	partial	yes
7	partial	partial	yes
8	partial	partial	yes
9	accepted	accepted	yes
10	accepted	accepted	yes
11	partial	partial	yes
12	accepted	accepted	yes
13	partial	partial	yes
14	accepted	accepted	yes
15	not_addressed	not_addressed	yes
16	not_addressed	not_addressed	yes
17	partial	partial	yes

	classification_match \
0	True
1	True
2	True
3	True
4	True
5	True
6	True
7	True
8	True
9	True
10	True
11	True
12	True
13	True
14	True
15	True
16	True
17	True

comment \

- 0
Easy exact-numbered match with clear agreement language.
- 1
Mixed disagreement plus explanation is treated as partial rather than simple rejection.
- 2
Known limitation: nearby repeated response headings make exact numbering alone fragile, although the final row is usable.
- 3
Clear partial acceptance language with qualifications.
- 4
Explicit rejection example.

5
 Structured response match with exploratory commitment language.
 6 Hard case where
 section-heading cleanup avoids recommendation-text leakage.
 7
 Structured ordinal response match.
 8 Good example of
 partial implementation against a multi-part recommendation.
 9
 Exact label match with clear agreement language.
 10
 Grouped/nested response case; label preservation is important.
 11
 Grouped response with mixed commitments across jurisdictions.
 12
 Clear accepted response in a long inquiry document.
 13
 Qualified support is correctly not treated as full acceptance.
 14 Hard case from embedded
 inquiry recommendations; final row is correctly aligned.
 15 Not-addressed example caused
 by responsibility outside UK government response scope.
 16
 Second not-addressed example with exact numbering.
 17 Select-committee
 extraction case; conclusion-only items are excluded upstream.

suggested_fix_if_error

0 None -
 correct in sample.
 1 Keep as a regression case for disagreement-with-
 alternative wording.
 2 Add stronger heading disambiguation and keep this as a hard-case
 regression test.
 3 None -
 correct in sample.
 4 None -
 correct in sample.
 5 None -
 correct in sample.
 6 Keep cleanup checks for trailing headings in
 future exports.
 7 None -
 correct in sample.
 8 None -
 correct in sample.
 9 None -
 correct in sample.
 10 Keep grouped-response unit tests for

```

nested labels.
11         Review jurisdiction-specific sublabels if a finer
benchmark is added.
12                                     None -
correct in sample.
13                                     None -
correct in sample.
14         Keep inline-recommendation extraction checks for
Grenfell-style prose.
15                                     None -
correct in sample.
16                                     None -
correct in sample.
17         Keep select-committee section-boundary checks for
future documents.

      label  sample_rows
0      partial           10
1      accepted           5
2  not_addressed          2
3      rejected           1

```

Task 2c representative sample rows: 18
Full-corpus accuracy/F1 is not reported because a complete manual benchmark is unavailable.

11. Examples And Hard Cases

These static tables keep representative evidence visible in PDF export.

```

example_keys = {
    ("behaviour_change", "8.1"): "accepted example",
    ("post_office", "4"): "partially accepted example",
    ("post_office", "13"): "rejected example",
    ("covid_inquiry_module2", "1"): "not addressed example",
    ("covid_inquiry_module2", "13"): "not addressed example",
}
example_df = final_df[final_df.apply(lambda row:
(str(row["document_pair"]), str(row["id"])) in example_keys,
axis=1)].copy()
example_df["evidence_note"] = example_df.apply(lambda row:
example_keys[(str(row["document_pair"]), str(row["id"]))], axis=1)
example_df["recommendation_preview"] =
example_df["recommendation_text"].fillna("").str.slice(0, 150) + "..."
example_df["response_preview"] =
example_df["matched_response_text"].fillna("").str.slice(0, 150) +
"..."
display(example_df[["evidence_note", "document_pair", "id",
"classification", "confidence", "debug.alignment_method",

```

```

"debug.alignment_confidence", "debug.classification_confidence",
"recommendation_preview",
"response_preview"]].sort_values(["document_pair", "id"]))

hard_notes = {
    ("behaviour_change", "8.33"): "response heading repeats 8.32, so
exact numbering is insufficient",
    ("space_economy", "30"): "trailing section-heading cleanup
prevents recommendation leakage",
    ("blood_inquiry", "4a i"): "nested labels are preserved for
grouped response structure",
    ("grenfell_phase2", "58"): "inline/fallback extraction handles
embedded inquiry recommendations",
    ("summer_2024_disorder", "1"): "select-committee extraction
excludes conclusion-only items",
}
hard_df = final_df[final_df.apply(lambda row:
(str(row["document_pair"]), str(row["id"])) in hard_notes,
axis=1)].copy()
hard_df["case_note"] = hard_df.apply(lambda row:
hard_notes[(str(row["document_pair"]), str(row["id"]))], axis=1)
hard_df["recommendation_preview"] =
hard_df["recommendation_text"].fillna("").str.slice(0, 150) + "..."
hard_df["response_preview"] =
hard_df["matched_response_text"].fillna("").str.slice(0, 150) + "..."
display(hard_df[["case_note", "document_pair", "id", "classification",
"confidence", "debug.alignment_method", "recommendation_page",
"matched_response_page", "recommendation_preview",
"response_preview"]].sort_values(["document_pair", "id"]))

```

	evidence_note	document_pair	id
classification \			
0	accepted example	behaviour_change	8.1
accepted			
218	not addressed example	covid_inquiry_module2	1
not_addressed			
230	not addressed example	covid_inquiry_module2	13
not_addressed			
45	rejected example	post_office	13
rejected			
36	partially accepted example	post_office	4
partial			

	confidence	debug.alignment_method	debug.alignment_confidence \
0	0.95	exact_label	0.95
218	0.85	exact_label	0.95
230	0.85	exact_label	0.95
45	0.95	exact_label	0.95
36	0.95	exact_label	0.95

```
debug.classification_confidence \
0 0.78
218 0.90
230 0.90
45 0.85
36 0.80
```

```
recommendation_preview \
```

```
0 The idea of the Government intervening to change people's
behaviour will often be controversial, and so it is important that
ministers a...
```

```
218 Recommendation 1: Chief Medical Officer for Northern Ireland The
Department of Health (Northern Ireland) should reconstitute the role
of...
```

```
230 Recommendation 13: Amendment of the Ministerial Code in
Northern Ireland The Executive Office should amend the Ministerial
Code to impos...
```

```
45 The current Dispute Resolution Procedure in HSS should be closed
once all claimants currently within the Procedure have either (a)
settl...
```

```
36 All claimants in HSS shall be entitled to obtain legal advice
funded by the Department prior to choosing between accepting the Fixed
Sum...
```

```
response_preview
```

```
0 The Government agrees with the Committee's conclusion. It is
important that behavioural interventions draw from the available
evidence b...
```

```
218
```

```
This recommendation is not for the UK government to respond to....
```

```
230
```

```
This recommendation is not for the UK government to respond to....
```

```
45 DBT rejects this recommendation as we believe it comes into
conflict with the principle of "full and fair" redress set out as we
accepte...
```

```
36 DBT broadly accepts this recommendation. DBT accepts that some
HSS claimants require access to legal support on their decision to
take t...
```

```
case_note \
```

```
32 response heading repeats 8.32, so exact numbering is
insufficient
```

```
109 nested labels are preserved for grouped response
structure
```

```
217 inline/fallback extraction handles embedded inquiry
recommendations
```

```
66 trailing section-heading cleanup prevents recommendation
```

leakage
237 select-committee extraction excludes conclusion-only
items

	document_pair	id	classification	confidence	\
32	behaviour_change	8.33	partial	0.95	
109	blood_inquiry	4a i	accepted	0.90	
217	grenfell_phase2	58	accepted	0.95	
66	space_economy	30	partial	0.95	
237	summer_2024_disorder	1	partial	0.95	

	debug.alignment_method	recommendation_page	matched_response_page	\
32	exact_label	72	18	
109	structured_grouped	288	30	
217	exact_label	248	30	
66	structure	107/108	16	
237	exact_label	38	4	

recommendation_preview \

32 We recommend that the Government (a) establish and publish targets for a reduction in carbon emissions as a result of a reduction in car...

109 A statutory duty of candour in healthcare should be introduced in Northern Ireland....

217 that the advice contained in paragraph 79.11 of the LGA Guide be reconsidered....

66 If the Government wishes to progress with the pursuit of sovereign launch capability, it should consider designating UK spaceports as Cr...

237 Notwithstanding potential changes to contempt of court laws, we recommend that the CPS publish its new media protocol as soon as possibl...

response_preview

32 The Government does not believe that it is appropriate to set specific targets for reducing carbon emissions by reducing car use. The Go...

109 This recommendation is accepted in full by the Northern Ireland Executive This recommendation is accepted in full by the Scottish Govern...

217 The government accepts this recommendation. The advice contained in paragraph 79.11 of the LGA Guide was redacted in 2021. The Home Offi...

66 The Government is committed to securing assured access to space for the UK, which requires access to both a spaceport and launch vehicle...

237 The CPS supported the committee to scrutinise the process by which information was shared with the public following the appalling attack...

12. Optional Interactive Extraction And Alignment Demo

Interactive widgets are optional. The static evidence above is sufficient for PDF marking: it includes the final 246-row summary, pair counts, classification distribution, validation checks, the Task 2c expected-vs-actual sample, representative examples and hard cases. When `ipywidgets` is unavailable or a notebook viewer cannot render widget state, the cells below also provide static fallback tables.

```
LABEL_COLOURS = {
    "accepted": "#198754",
    "partial": "#fd7e14",
    "partially_accepted": "#fd7e14",
    "rejected": "#dc3545",
    "not_addressed": "#6c757d",
}

try:
    from src.widgets.extraction_explorer import show as
    show_extraction_explorer

    display(Markdown("Interactive extraction explorer loaded. If your
notebook viewer cannot render widgets, use the static fallback table
below.))
    show_extraction_explorer(demo_recommendations, demo_alignments,
classify_response, LABEL_COLOURS)
except Exception as exc:
    display(Markdown(
        f"Interactive extraction explorer unavailable in this
environment (`{type(exc).__name__}`: {exc})). "
        "Static demo alignment rows are shown below for marking."
    ))

fallback_alignment_df = pd.DataFrame(demo_alignments)
if fallback_alignment_df.empty:
    fallback_alignment_df = pd.DataFrame(demo_recommendations)
preview_cols = [
    col for col in [
        "recommendation_id",
        "id",
```

```

        "response_id",
        "classification",
        "confidence",
        "alignment_confidence",
        "method",
        "recommendation_text",
        "matched_response_text",
    ]
    if col in fallback_alignment_df.columns
]
display(fallback_alignment_df[preview_cols].head(10) if preview_cols
else fallback_alignment_df.head(10))

```

Optional Final-Row Inspector

This small inspector is separate from the widget above. It lets a live notebook user choose any final export row and view the recommendation, matched response, classification, confidence and provenance. If widgets are unavailable, the cell displays a static sample of final export rows instead.

```

import html
from IPython.display import HTML, display

final_export_df = final_df.reset_index(drop=True).copy()

def _row_label(row: pd.Series) -> str:
    return f"{row['document_pair']} {row['id']} - {LABEL_DISPLAY.get(row['classification'], row['classification'])}"

def _render_final_row(row_index: int) -> HTML:
    row = final_export_df.iloc[int(row_index)]
    recommendation = html.escape(str(row.get('recommendation_text', '')) or '')
    response = html.escape(str(row.get('matched_response_text', '') or ''))
    classification =
html.escape(str(LABEL_DISPLAY.get(row.get('classification'), row.get('classification'))))
    confidence = html.escape(str(row.get('confidence', '')))
    align_method = html.escape(str(row.get('debug.alignment_method', '')))
    align_conf = html.escape(str(row.get('debug.alignment_confidence', '')))
    response_page = html.escape(str(row.get('matched_response_page', '')))
    row_id = html.escape(f"{row.get('document_pair')} {row.get('id')}")

```

```

    return HTML(f"""
        <div style="border:1px solid #ddd;padding:10px;border-
radius:6px;font-family:sans-serif;">
            <div><strong>Recommendation ID:</strong> {row_id}</div>
            <div><strong>Classification:</strong> {classification}</div>
            <div><strong>Confidence:</strong> {confidence}</div>
            <div><strong>Alignment method:</strong> {align_method}
(confidence {align_conf})</div>
            <div><strong>Matched response page:</strong>
{response_page}</div>
            <hr style="border:none;border-top:1px solid #eee;margin:8px
0;"/>
            <div><strong>Recommendation text</strong></div>
            <div style="max-height:180px;overflow:auto;white-space:pre-
wrap;">{recommendation}</div>
            <div style="margin-top:8px;"><strong>Matched response
text</strong></div>
            <div style="max-height:220px;overflow:auto;white-space:pre-
wrap;">{response}</div>
            </div>
        """)

try:
    import ipywidgets as widgets

    row_dropdown = widgets.Dropdown(
        options=[(_row_label(row), idx) for idx, row in
final_export_df.iterrows()],
        value=0,
        description="Row",
        layout=widgets.Layout(width="100%"),
    )
    row_output = widgets.Output()

    def _update_row(change=None) -> None:
        with row_output:
            row_output.clear_output(wait=True)
            display(_render_final_row(row_dropdown.value))

    row_dropdown.observe(_update_row, names="value")
    _update_row()
    display(widgets.VBox([row_dropdown, row_output]))
except Exception as exc:
    display(Markdown(
        f"Final-row widget unavailable in this environment
(`{type(exc).__name__}`: {exc}"). "
        "A static sample of final export rows is shown below."
    ))
    static_cols = [
        "document_pair",

```

```

        "id",
        "classification",
        "confidence",
        "debug.alignment_method",
        "debug.alignment_confidence",
        "recommendation_text",
        "matched_response_text",
    ]
    static_cols = [col for col in static_cols if col in
final_export_df.columns]
    display(final_export_df[static_cols].head(10))

```

13. Challenges, Optional Web Interface And Limitations

Challenges and resolutions. The workflow handles unreliable numbering, varied recommendation layouts, quoted recommendation text inside responses, nested labels and incomplete manual ground truth. It uses documented preset-specific settings where needed and preserves method/confidence metadata so hard cases can be inspected.

Automated QA evidence. The `tests/` folder contains regression checks for response-unit extraction, structured matching, classification corrections, final export validation and the frontend evaluation-summary helpers. These tests were used to repeatedly check known hard cases while refining the extraction/alignment pipeline.

Optional web interface. The FastAPI/Vite interface exposes the same project workflow interactively, with screenshots in `screenshots/` as supporting evidence. The final submission is local-first because the full corpus, optional model setup and pipeline outputs are better suited to local marking than limited free hosting.

Limitations and future work. Add full 246-row manual labels for accuracy/F1, keep expanding hard-case tests for new document layouts, improve OCR quality auditing and consider a second-stage semantic reranker for weak matches after submission.

14. Final Conclusion

The final Task 2 evidence shows 246 recommendation-level rows across the full corpus of eight document pairs / 16 PDFs, with the expected pair counts and classification distribution. It covers Task 2a recommendation extraction, Task 2b response matching/classification and Task 2c evaluation through validation checks plus a representative expected-vs-actual sample, while avoiding any claim of full-corpus manual accuracy. The notebook uses the shared project modules, preserves confidence metadata, references the final JSON/CSV exports, provides static and optional interactive evidence, and avoids overwriting final outputs.